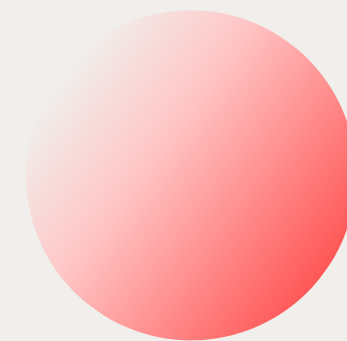


TRACK ONE

BIG DATA & AI BOOTCAMP

MongoDB Project



Desert Ninjas

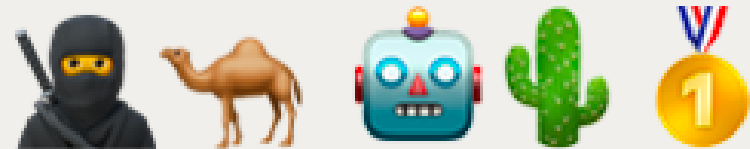
Table of Content



- * Team members
- * Volcanos of earth dataset analysis
- * H&M dataset analysis

Team Members

DESERT NINJAS



Reema alaswad
Maha Alhazzani
Aljohara alkanhal
Raghad Aleisa
Eman aldosari

Volcano of the earth

FIRST DATASET

20 countries that have had volcanoes and,
there are **2,662 erupted volcanoes**

Contains information about the volcanos such as the
name, type, image, country, latitude, longitude,
population.

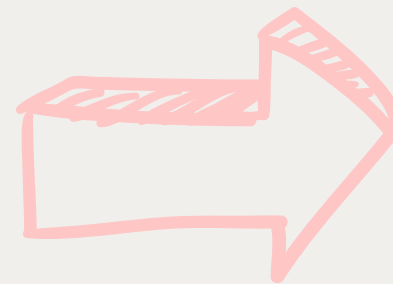


Pipeline One

FIRST DATASET

- Get all volcanoes that located in Japan, with the fields "Volcano_Name", "Volcano_Type", "Country", "Last_Eruption", "Lat", "Lon"

```
[{ $project: { _id: 0,  
  Volcano_Name: 1,  
  Volcano_Type: 1,  
  Country: 1,  
  Last_Eruption: 1,  
  Lat: 1,  
  Lon: 1 }},  
{ $match: {  
  Country: {  
    $eq: 'Japan' } }},  
{ $out: 'AllVolcanoesInJapan' }]
```



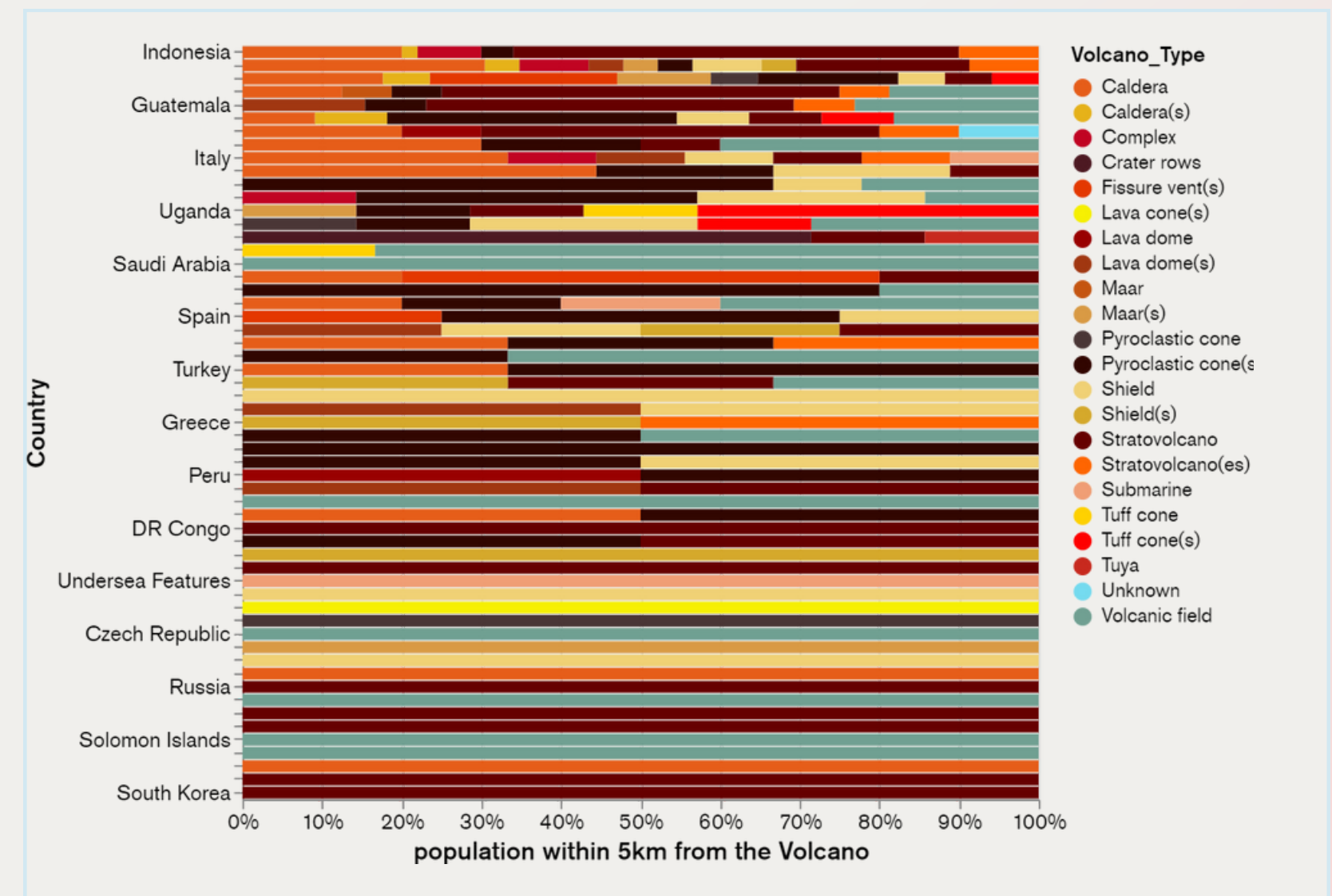
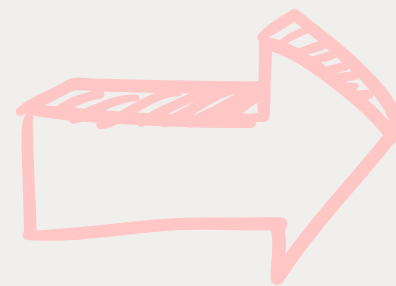
Volcanoes in Japan Mean Latitude and Longitude			
Mean of Latitude and Longitude for each Volcano that have occurred in Japan			
Volcano_Name	mean (Lat)	mean (Lon)	count (Volc...
Abu	34.5	131.6	1
Adachi	38.218	140.654	1
Adatarayama	37.647	140.281	1
Aira	31.593	130.657	1
Aizu Nunobiki	37.321	140.018	1
Akagisan	36.56	139.193	1
Akaigawa	43.08	140.82	1
Akan	43.384	144.013	1
Akandanayama	36.2	137.573	1
Akita-Komagatake	39.761	140.799	1
Akita-Yakeyama	39.964	140.757	1
Akusekijima	29.465	129.594	1

Pipeline Two

FIRST DATASET

- Get all volcanoes where population within 5 km more than 5 thousand people, with the fields "Volcano_Name", "Volcano_Type", "Country", "Last_Eruption", "Lat", "Lon", "population_within_5km"

```
[{ $project: { _id: 0,
  Volcano_Name: 1,
  Volcano_Type: 1,
  Country: 1,
  Last_Eruption: 1,
  Lat: 1,
  Lon: 1,
  population_within_5km: 1 },
  { $match: { population_within_5km: {
    $gt: 5000 } }},
  { $out:
    'AllVolcanoesWithPW5KMisMoreFiveK' } ]
```

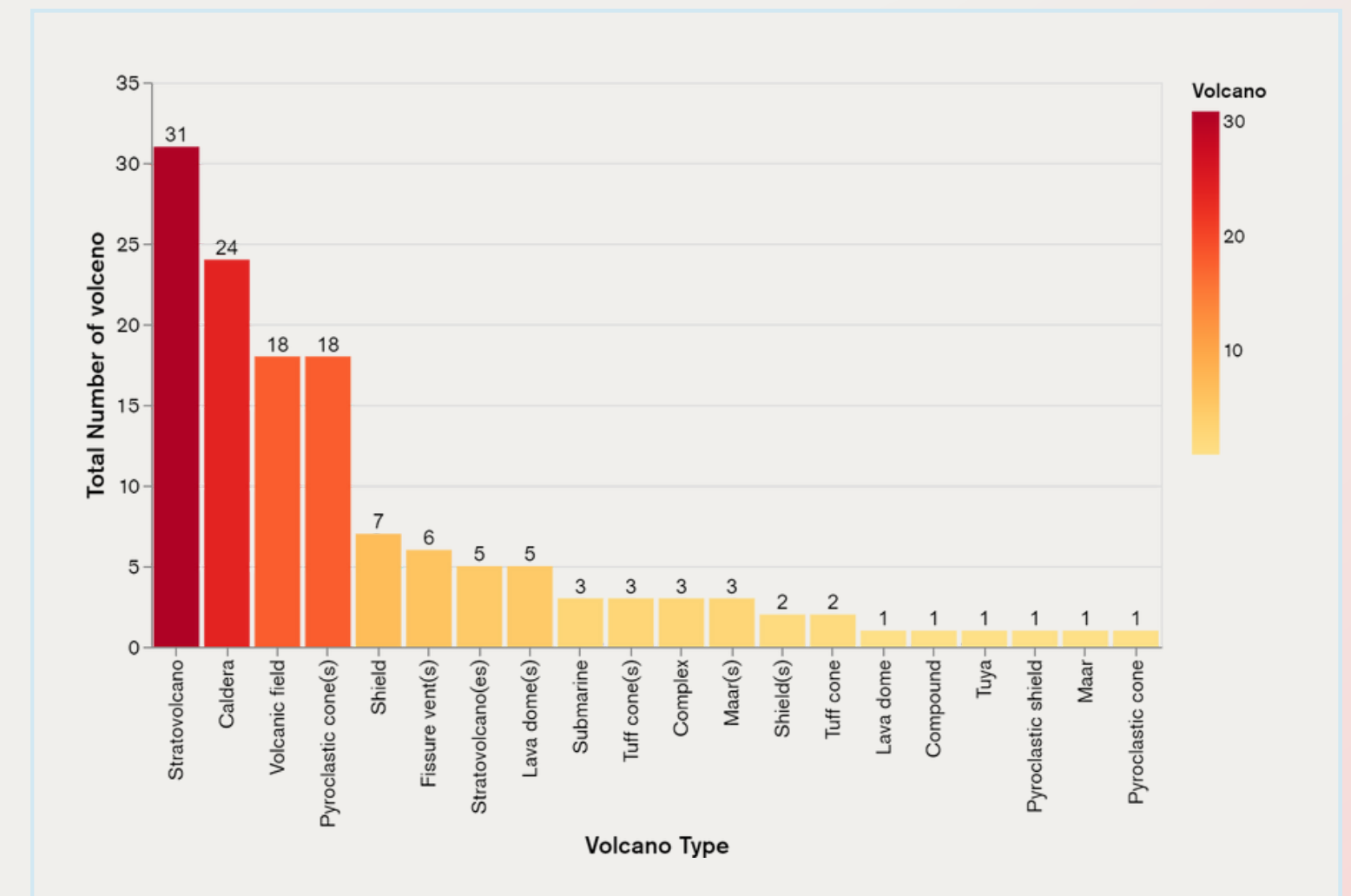
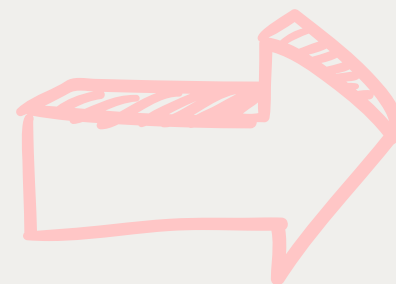


Pipeline Three

FIRST DATASET

- Get volcanoes per country per volcano type where the population is more than 100,000 people within 10km

```
[[{ $project: { _id: 0,
  Volcano_Name: 1, Volcano_Type: 1,
  Country: 1, population_within_10km: 1 }},
{ $match: {
  population_within_10km: { $gt: 100000 } }},
{ $group: { _id: '$Country',
  Grouping: { $push: { Volcano_Name:
'$Volcano_Name', Volcano_Type:
'$Volcano_Type', population_within_10km:
'$population_within_10km' } }},
  Total_Of_Volcanos: { $sum: 1 } }},
{ $out:
'VolcanoesPerCountryPerVolcanoTypePW10KM' } ]]
```

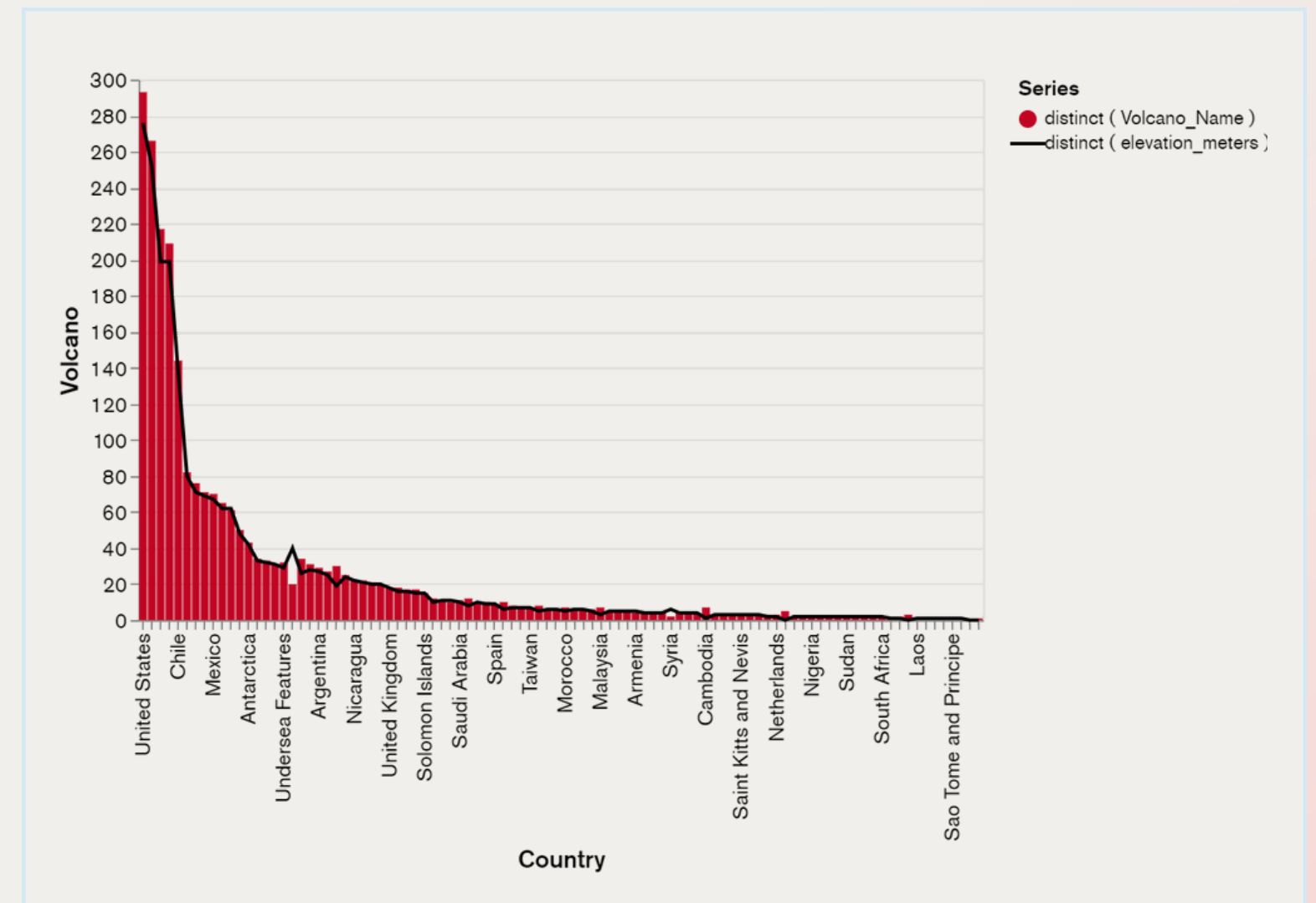
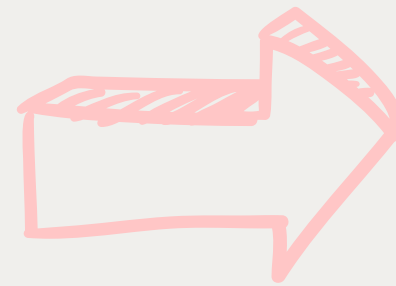


Pipeline Four

FIRST DATASET

- Retrieve all volcanoes that erupted before the common era (BCE), and get the name, location (country, lon, lat), and elevation. Sort by elevation level from highest to lowest (some of them underwater)

```
[{ $project: { _id: 0,
  Volcano_Name: 1,
  Country: 1,
  Lat: 1, Lon: 1,
  Last_Eruption: 1,
  elevation_meters: 1}},
{ $match: {
  Last_Eruption: {
    $not: RegExp("BCE") } }},
{ $sort: {
  elevation_meters: -1 }},
{ $out: 'AllVolcanoesEruptedBeforeBCE'}]
```

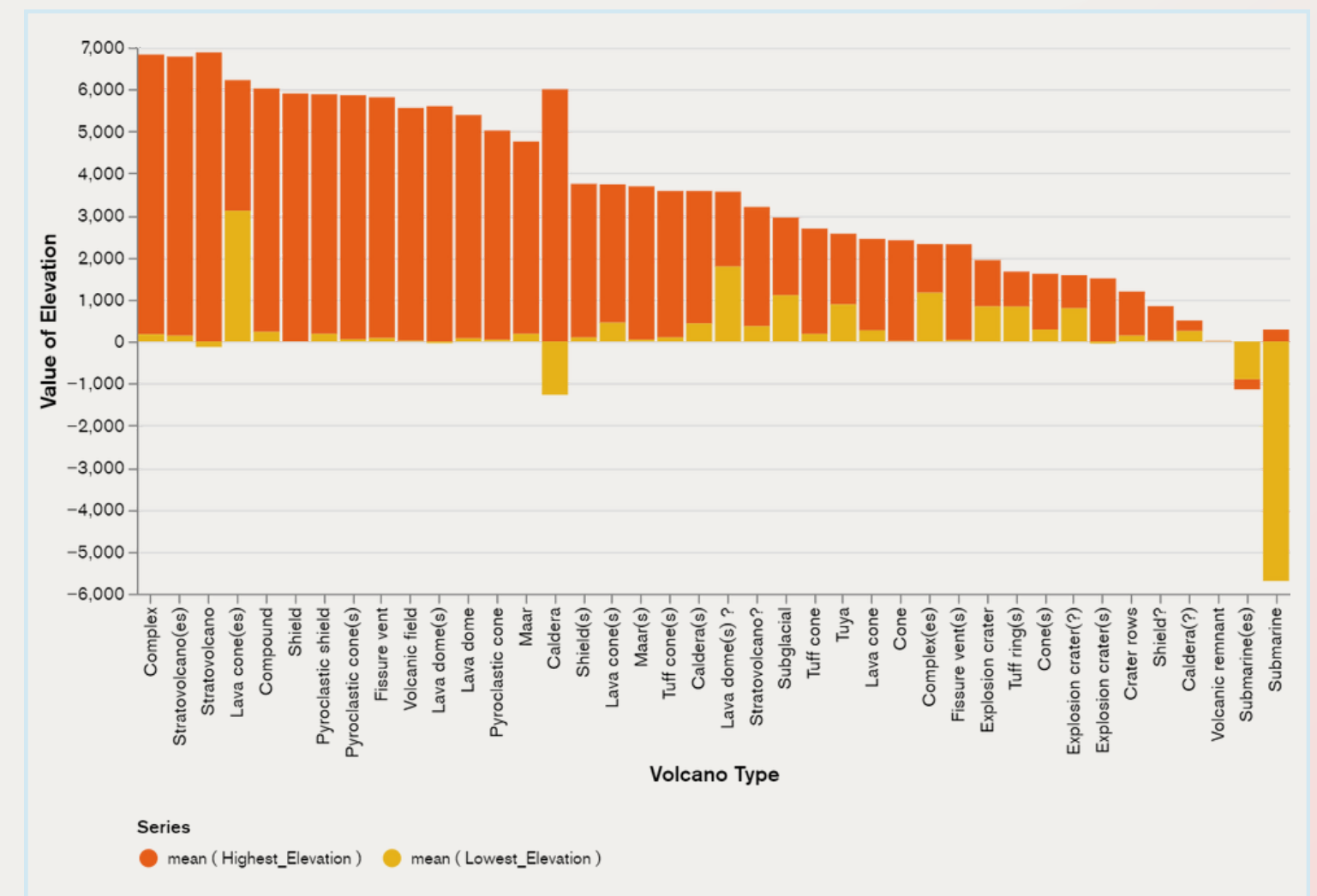
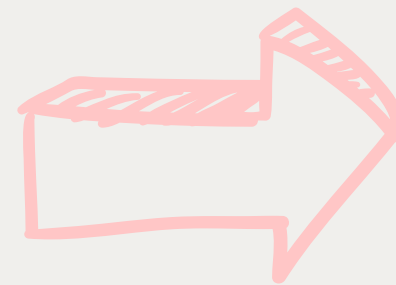


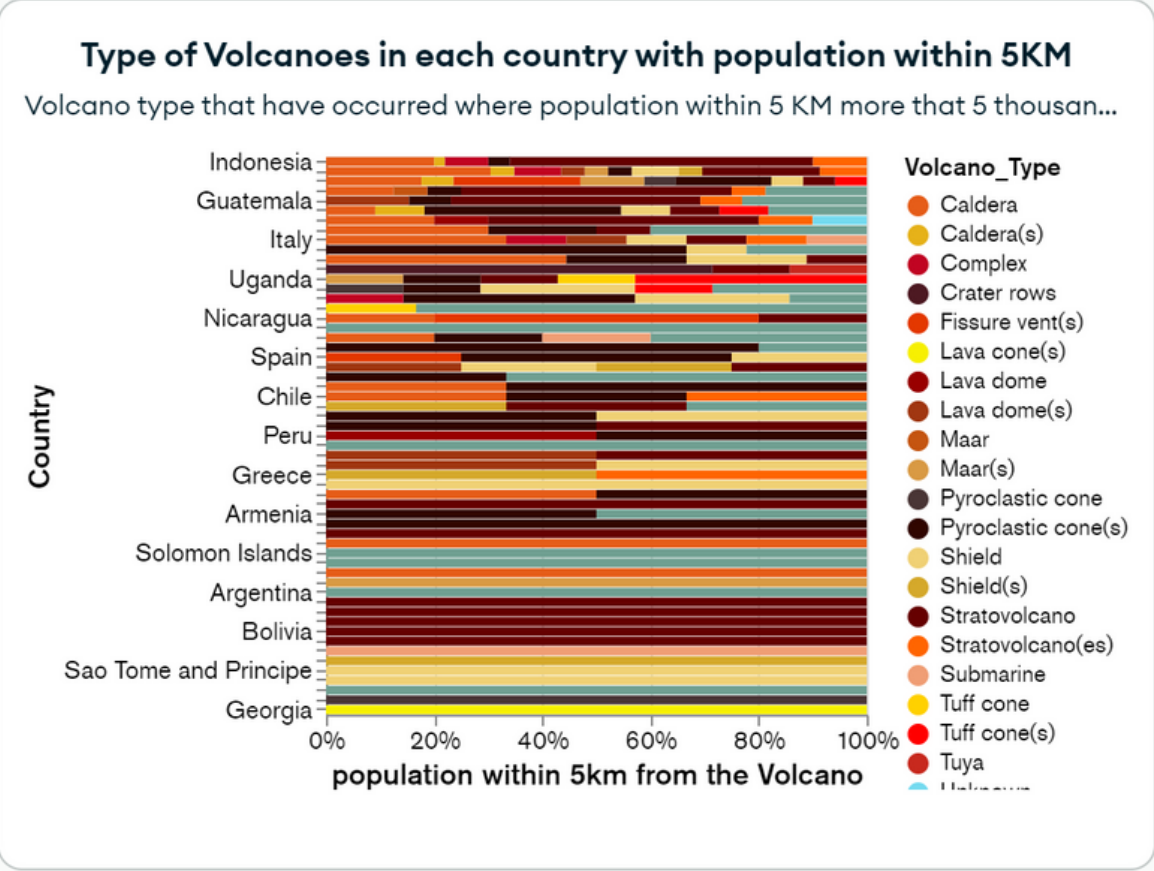
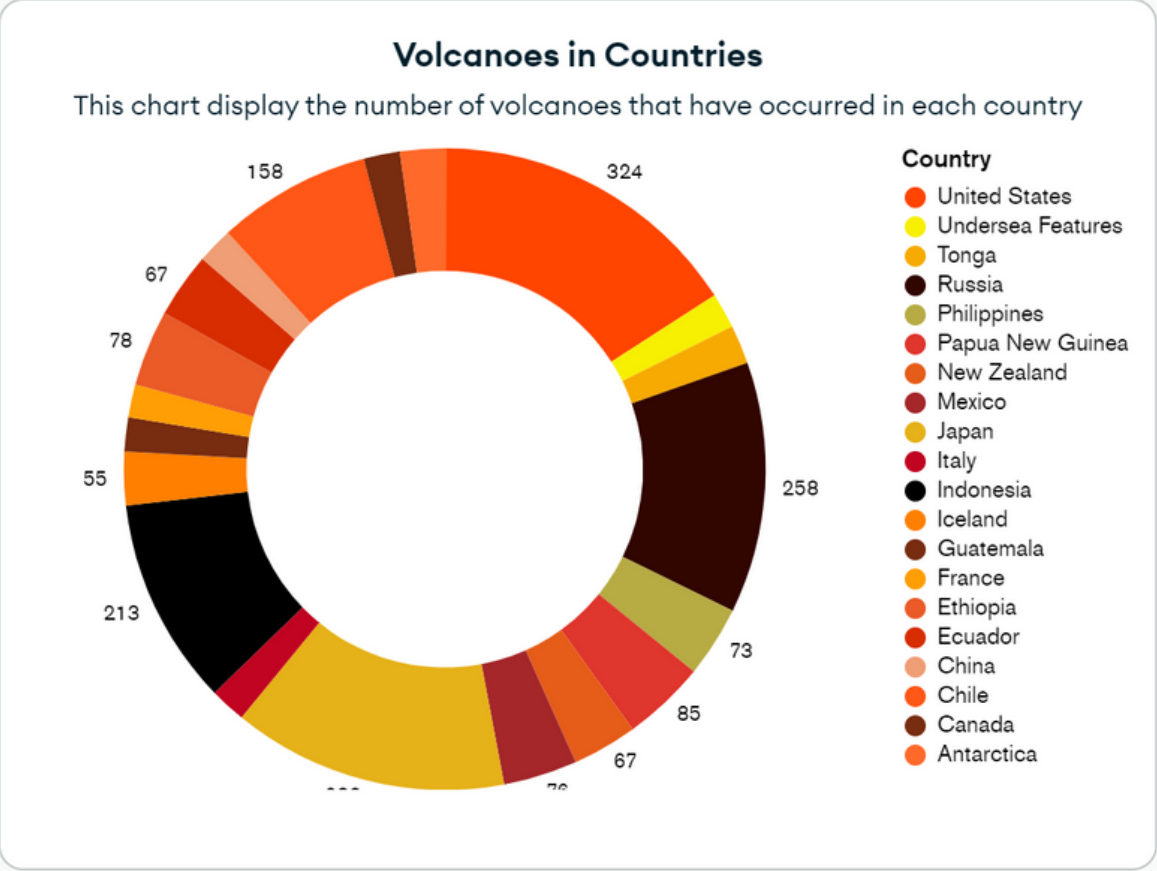
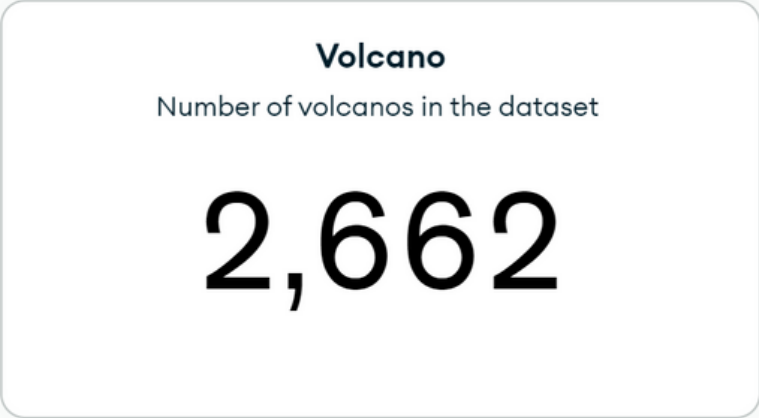
Pipeline Five

FIRST DATASET

- Get volcanoes with the highest and lowest elevation by volcano types.

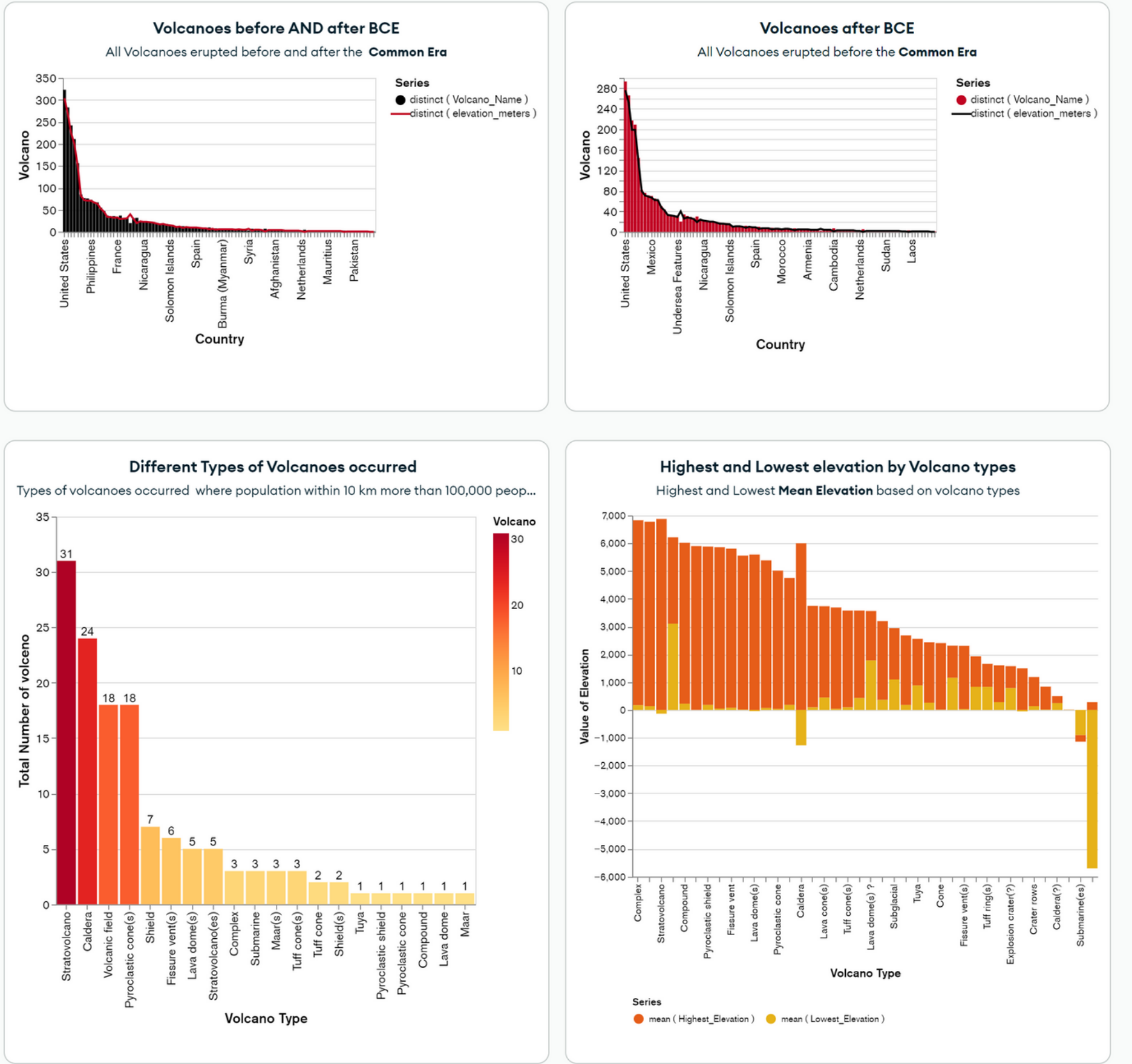
```
{ { $project: { _id: 0,  
Country: 1,  
Volcano_Name: 1,  
Volcano_Type: 1,  
elevation_meters: 1 } },  
{ $group: { _id: '$Volcano_Type',  
Highest_Elevation: {  
$max: '$elevation_meters',  
Lowest_Elevation: {  
$min: '$elevation_meters' } } },  
$out:  
'HighestAndLowestElevationByVolcanoTypes' ] }
```

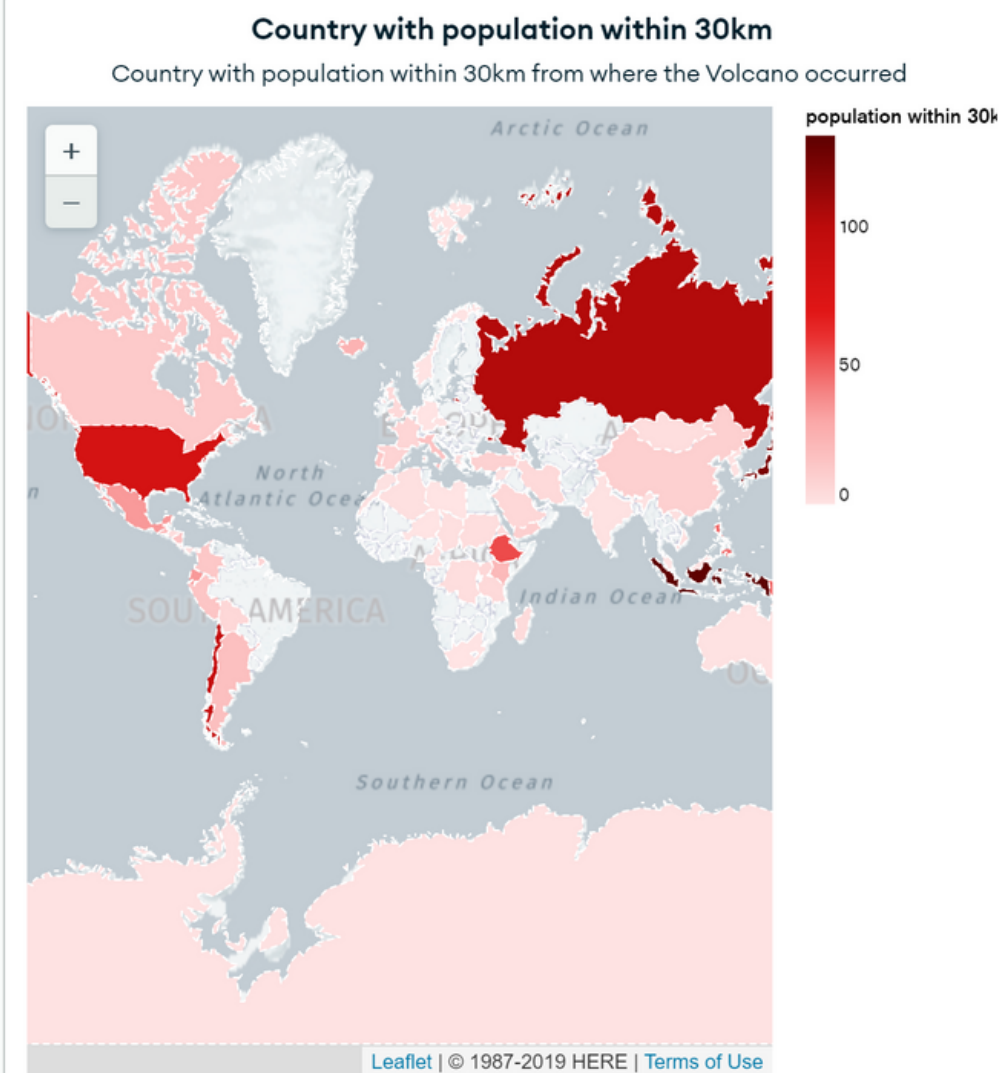




DASHBOARD

FIRST DATASET





Volcanoes in Japan Mean Latitude and Longitude
Mean of Latitude and Longitude for each Volcano that have occurred in Japan

Volcano_Name	mean (Lat)	mean (Lon)	coi
Abu	34.5	131.6	
Adachi	38.218	140.654	
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Aira	31.593	130.657	
Aizu Nunobiki	37.321	140.018	
Akagisan	36.56	139.193	
Akaigawa	43.08	140.82	
Akan	43.384	144.013	
Akandanayama	36.2	137.573	
Akita-Komagatake	39.761	140.799	
Akita-Yakeyama	39.964	140.757	

H&M DATASET ANALYSIS



H&M DATASET Overview

SECOND DATASET

- Contains information about the products such as the name, type, color, description, price.
- There are **+100k** Documents



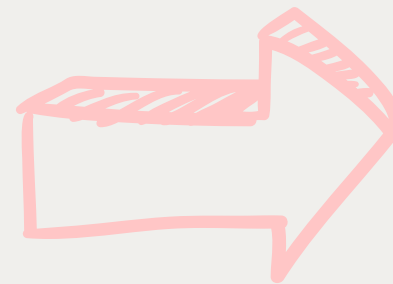
Functions

Function One

SECOND DATASET

- A function that returns all the *red or orange t-shirts*.

```
{ $match: {  
  product_type_name: {  
    $eq: 'T-shirt'  
  },  
  colour_group_name: {  
    $in: [ 'Red', 'orange' ]  
  }  
}}
```



```
# Demo how function works  
# check length of the listings  
len(Return_Red_OR_Orange['response'])
```

231

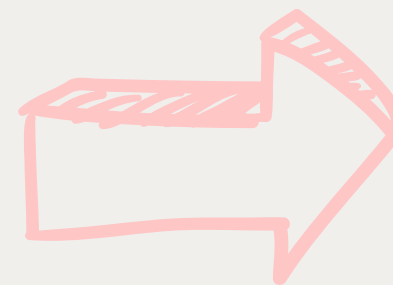
```
[{'_id': ObjectId('634a8fd09aeff8deb6d3b3ba'),  
  'article_id': '265071030',  
  'product_code': 265071,  
  'prod_name': 'KRISTOFF Basic jersey polo BB',  
  'product_type_no': 255,  
  'product_type_name': 'T-shirt',  
  'product_group_name': 'Garment Upper body',  
  'graphical_appearance_no': 1010016,  
  'graphical_appearance_name': 'Solid',  
  'colour_group_code': 42,  
  'colour_group_name': 'Red',  
  'perceived_colour_value_id': 7,  
  'perceived_colour_value_name': 'Medium',  
  'perceived_colour_master_id': 18,  
  'perceived_colour_master_name': 'Red',  
  'department_no': 8768,  
  'department_name': 'Young Boy Jersey Basic',  
  'index_code': 'I',  
  'index_name': 'Children Sizes 134-170',  
  'index_group_no': 4,  
  'index_group_name': 'Baby/Children',
```

Function Two

SECOND DATASET

- A function that takes two arguments for *index_group_name* and *colour_group_name* and returns all documents and it is *case insensitive*.

```
return collection.find({
  index_group_name: { '$regex': arg1, "$options": "i" },
  colour_group_name: { '$regex': arg2, "$options": "i" } })
```



```
[ ] # Demo how function works
# check length of the listings
len(Return_Index_Color['response'])
```

679

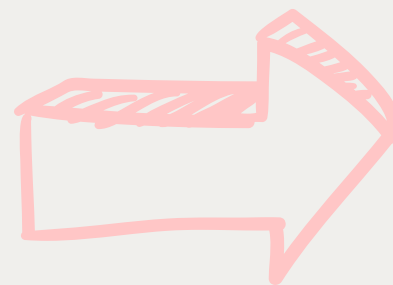
```
[{'_id': ObjectId('634a8fd09aeff8deb6d3b345'),
  'article_id': '245348002',
  'product_code': 245348,
  'prod_name': 'Santa Hat',
  'product_type_no': 75,
  'product_type_name': 'Hat/beanie',
  'product_group_name': 'Accessories',
  'graphical_appearance_no': 1010014,
  'graphical_appearance_name': 'Placement print',
  'colour_group_code': 42,
  'colour_group_name': 'Red',
  'perceived_colour_value_id': 7,
  'perceived_colour_value_name': 'Medium',
  'perceived_colour_master_id': 18,
  'perceived_colour_master_name': 'Red',
  'department_no': 9985,
  'department_name': 'Accessories',
  'index_code': 'F',
  'index_name': 'Menswear',
  'index_group_no': 3,
  'index_group_name': 'Menswear',
  ...}]
```

Function Three

SECOND DATASET

- Create a function that takes three arguments for product_type_name, colour_group_name, and price range (make it an array [...]), and returns the result with product_type_name, colour_group_name, price, department_name, and discount_%. String arguments have to be case-insensitive.

```
return collection.find({
  product_type_name: {"$regex": product_type, "$options": "i"},
  colour_group_name: {"$regex": colour_group, "$options": "i"},
  price: {"$gte": price_range_min, "$lte": price_range_max} },
{ id:0, product_type_name:1, colour_group_name:1,
price: 1, department_name:1, 'discount%':1} )
```



```
[ ] # Create a function that takes three arguments for product_type_name, colo

ProductDiscount = h_and_m.command("callFunction",
                                  "ProductDiscount",
                                  arguments=['vest top','black', 30, 50])
```

```
[ ] # Demo how function works
# check length of the listings
len(ProductDiscount['response'])
```

70

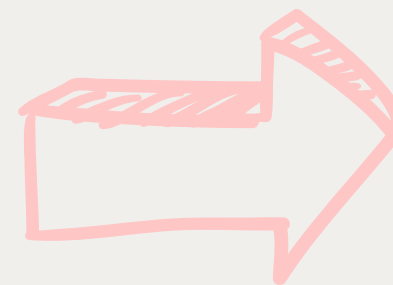
```
[ ] [ {'product_type_name': 'Vest top',
      'colour_group_name': 'Black',
      'department_name': 'Men Sport Bottoms',
      'price': 37.35,
      'discount_%': 25},
      {'product_type_name': 'Vest top',
      'colour_group_name': 'Black',
      'department_name': 'Jersey Fancy DS',
      'price': 42.42,
      'discount_%': 0},
      {'product_type_name': 'Vest top',
      'colour_group_name': 'Black',
      'department_name': 'Jersey Basic',
      'price': 37.85,
      'discount_%': 25},
      {'product_type_name': 'Vest top',
      'colour_group_name': 'Black',
      'department_name': 'Men Sport Bottoms',
      'price': 42.54,
      'discount_%': 15},
```

Function Four

SECOND DATASET

- A function that takes an argument for the *product type name*, calculates the discount for that product returns the product name, old price (before discount), discount, new price (after discount), and product description. Sort by new price from cheap to expensive. Limit to the first 50 results.

```
{ $match: {
  product_type_name: {
    $eq: arg } }, {
  $project: {
    _id: 0,
    prod_name: 1, price: 1, discount: '$discount_%',
    calculated_discount: { $multiply: [{ $divide: [
      '$discount_%', 100 ] }, '$price' ] }, detail_desc: 1 }, {
  $addFields: { new_price: { $round: [
    { $subtract: [
      '$price', '$calculated_discount'
    ] }, 2 ] } } }, {
  $sort: {
    new_price: 1 }, {
  $limit: 50 }
```



```
[ ] # Demo how function works
    # check length of the listings
    len(Calculate_Discount['response'])
```

50

```
[{'prod_name': 'Tulip LS FANCY 2PACK (TVP)',
  'detail_desc': 'Long-sleeved tops in soft, organic cotton jersey.',
  'price': 5.21,
  'discount': 20,
  'calculated_discount': 1.042,
  'new_price': 4.17},
 {'prod_name': '2P SS T-shirt mixed',
  'detail_desc': 'T-shirts in soft, organic cotton jersey.',
  'price': 5.76,
  'discount': 25,
  'calculated_discount': 1.44,
  'new_price': 4.32}]
```

Indexes

Index One

SECOND DATASET

- Creating a *Text Index* for all text fields

```
collection.create_index([("$**", pymongo.TEXT)])  
  
for review in collection.find({ "$text": { "$search": "black" } }):  
    print(review)
```

Index Two

SECOND DATASET

- Create a Compound Index for the *product type name* and *color*

```
collection.create_index([("product_type_name", pymongo.DESCENDING), ("colour_group_name",  
pymongo.ASCENDING)])
```

```
for review in collection.find({ "product_type_name": { "$regex": "top", "$options": "i" }, "colour_group_name":  
{ "$regex": "black", "$options": "i" } }):
```

```
    print(review)
```


Index Three

SECOND DATASET

- Create an index to improve the performance of the query to retrieve *product type*, *color*, and *price*

```
collection.create_index([("price", pymongo.ASCENDING)])
```

```
for review in collection.find({ "product_type_name": { "$regex": "Sweater", "$options": "i" },  
    "colour_group_name": { "$regex": "Off White", "$options": "i"}, "price": { "$lte": 100} }):  
    print(review)
```

DESERT NINJAS

Thanks for listening!

